

A Comparative Analysis of OCR Models for Key Information Extraction from Invoices

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Abstract:

Extracting structured data from scanned invoices is a growing need across digital operations. Traditional OCR (Optical Character Recognition) systems, although widely used, often fail to extract text cleanly and meaningfully in noisy or skewed images. This research presents a preprocessing-enhanced OCR pipeline, combined with a Named Entity Recognition (NER) model, to benchmark OCR accuracy based on downstream tasks. Using EasyOCR, PyTesseract, and PaddleOCR, we evaluate precision, recall, and F1-score through a fine-tuned RoBERTa token classifier. PaddleOCR achieved the highest F1-score of 0.958, outperforming the rest. Our work proposes an efficient method to evaluate OCR outputs not only on text accuracy but also on their ability to extract semantic information—a practical perspective for real-world automation systems.

KEYWORDS:

OCR, Token Classification, Invoice Automation, RoBERTa, NER, Deep Learning, EasyOCR, PaddleOCR, Financial Metadata Extraction, Natural Language Processing, Transformer Models, Real-world Document Processing, Handwritten Receipts, Financial Automation.

1. INTRODUCTION

Digitizing documents such as invoices is essential in sectors like finance, logistics, and enterprise resource management. Optical Character Recognition (OCR) plays a pivotal role in extracting textual data from these scanned inputs. However, standard OCR systems often struggle with misaligned documents, noise, faded text, and improper lighting. Moreover, while they retrieve

Text, they fall short of identifying meaningful entities such as companies, dates, and monetary values. This is where technology, more specifically, Optical Character Recognition (OCR), comes into play. OCR facilitates the conversion of scanned or photographed invoices into editable and searchable text. It's an effective tool; however, the problem is that text recognition alone is insufficient. [1] Companies need to understand not just what is written on the page, but also the meaning of each word and its context. For instance, consider an automated invoice processing system: It's not enough to know that the text contains the word "12/04/2023" or "John Smith." A smart system should also know that it's the invoice date, not a delivery date or due date. Similarly, identifying the name "John Smith" is less useful if the system can't tell whether he's the sender, recipient, or customer. Without this context, automating workflows such as expense tracking or vendor payments becomes significantly more challenging. This paper addresses these limitations by designing a two-step pipeline that first enhances image quality before applying OCR, and then evaluates extracted data through a token classification model. Unlike traditional benchmarks that measure OCR performance based solely on character-level or word-level accuracy, we go a step further by evaluating how well the OCR output supports Named Entity Recognition (NER)—a task with direct implications for financial automation. Our evaluation focuses on identifying key entities such as company names, dates, and personnel names, using a fine-tuned RoBERTa model. To validate our approach, we benchmark three popular OCR tools—PyTesseract, EasyOCR, and PaddleOCR—and assess them using a downstream token classification model. This not only reveals differences in raw text accuracy but also highlights which tools are better suited for semantic understanding. By shifting the focus from "what was

written" to "what was meant," we introduce a more practical and application-centric metric for OCR performance, which is particularly valuable in real-world, business-critical document processing. [2] The remainder of this paper is structured as follows: Section II outlines our proposed methodology, including preprocessing and model design. Section III details the implementation pipeline and training strategy. Section IV presents the evaluation results and discusses key insights. Finally, Section V concludes with future directions for expanding this framework to multilingual and mobile document processing.

II. PROPOSED METHODOLOGY

1) Pipeline Overview

Our system consists of two main components: an image preprocessing module and an evaluation phase. [3] The first component enhances the scanned image for OCR, while the second evaluates the extracted text semantically.

2) Image Preprocessing

Before feeding images into any OCR engine, we apply a carefully crafted preprocessing workflow, designed to enhance both text clarity and positional consistency. To improve OCR performance, each scanned invoice undergoes a sequence of operations:

1. **Barcode Masking** – Barcodes, if not handled properly, can introduce noise in OCR outputs as the engine might misinterpret them as text. To prevent this, we use the `mask_barcode_regions()` function to mask barcode regions, ensuring clean and meaningful text extraction.
2. **Orientation and Tilt Correction** – Receipts may be captured in horizontal or rotated formats. We standardize them into vertical alignment using the `orient_vertical()` function, which ensures consistency and better OCR accuracy.
3. **Edge Enhancement**—The `sharpen_edge()` function enhances text edges, making characters stand out from the background and improving clarity for OCR models.
4. **Binarization** – Using the `binarize()` function, we isolate text from noisy backgrounds, converting the image into high-contrast black-and-white to make characters clearer.
5. **Adaptive Thresholding** – Converts the image into a high-contrast format suitable for OCR

3) Text Extraction

Each cleaned image is passed through OCR models (PyTesseract, EasyOCR, and PaddleOCR). These

tools output both bounding box coordinates and recognized text. The format is: `[x1, y1, ..., x4, y4, "text"]`. This data is stored for the token classification phase.

III. IMPLEMENTATION & TRAINING

A. Entity Labeling Scheme

To evaluate semantic understanding, the extracted text is labeled using the **BIO (Beginning-Inside-Outside)** tagging scheme:

- B-COMPANY / I-COMPANY for organization names
- B-DATE / I-DATE for invoice dates
- B-PERSON for personnel
- O for unrelated tokens

The bounding box coordinates are normalized, and tokens are mapped to corresponding labels using a dictionary of ground truth annotations.

B. Model and Dataset

We utilize the RoBERTa base model, pre-trained on large text corpora, and fine-tune it for token classification. We used the publicly available **SROIE (Scanned Receipt OCR and Information Extraction) Dataset V2**, which comprises **1,000 scanned receipt images** from real-world business settings. The dataset is split into **626 training images** and **347 testing images**, annotated for four key fields: Company Name, Date, Address, and Total.

Diversity: The invoices vary in layout, fonts, logo positioning, and formatting (tabular and linear styles). However, the dataset contains only **English-language receipts**, which restricts multilingual generalization.

Annotation Quality: Labels are manually created with tight bounding boxes and accurate BIO tagging per token. Minor misalignments may occur due to OCR noise, typically from faded ink or poor scanning quality in a few samples.

C. Training Strategy

Model training is managed using the HuggingFace Trainer API with the following parameters:

- Learning rate: 3e-5
- Batch size: 16
- Epochs: 10
- Evaluation: Precision, Recall, F1-Score

The training loop also includes early stopping and dynamic learning rate scheduling.

Algorithm: Pre-processing and Data Extraction using various OCR models

- 1: **Input:** Scanned document image.
- 2: **Output:** Structured text data with bounding boxes.
- 3: **Pre-processing Phase**
- 4: MaskBarcodeRegions(image):
 - a. Convert to grayscale
 - b. Compute the gradient and apply a threshold
 - c. Use morphology to detect and mask the barcode
- 5: CorrectOrientation(image):
 - a. Rotate if width > height
- 6: EnhanceEdges(image):
 - a. Apply Gaussian blur and dilation
 - b. Detect edges using Canny
- 7: Binarize(image):
 - a. Threshold using image mean
 - b. Apply dilation to highlight text
- 8: DetectAndCorrectTilt(image):
 - a. Find the largest contour
 - b. Compute the angle and rotate
- 9: CropAndTrimBorders(image):
 - a. Crop to bounding box
 - b. Trim black borders
- 10: FinalEnhancement(image):
 - a. Convert to grayscale
 - b. Apply adaptive thresholding

11: Extraction Phase

- 12: ExtractAndStoreText(image):
 - a. Apply OCR (e.g., EasyOCR)
 - b. For each detected text:
 - Get bounding box (x1, y1, x2, y2, x3, y3, x4, y4)
 - Get text content
 - Store as: [x1, y1, ..., x4, y4, "text"]
 - 13: **Output:** Extracted text files.
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Fig 1: Input receipt image used for structured text extraction.



Fig 2: Output of the OCR system showing extracted structured fields annotated on the receipt image.

To make the OCR output compatible with layout-aware NLP models such as LayoutLM or RoBERTa, we define a preprocessing pipeline via the `preprocess_example()` function. [4] This function tokenizes the document, normalizes bounding box coordinates (scaled to a 0–1000 range), and assigns BIO labels by mapping annotations to the detected tokens. It also leverages image size metadata and HuggingFace processors to generate input features like `input_ids`, `pixel_values`, `attention_mask`, and `labels`. After preprocessing, all .jpg images from the dataset directory are batch-processed into this format. The results are compiled using `Dataset.from_dict()`, which neatly integrates the data into HuggingFace’s training and evaluation workflows.

D. Model Choice Justification

While layout-aware architectures such as LayoutLMv2, DocFormer, and TrOCR offer superior document understanding by leveraging both text and layout features, our goal was to assess the isolated impact of OCR model quality in a semantically-aware pipeline. We selected RoBERTa due to:

- Seamless integration with HuggingFace’s token classification pipelines
- Lower computational requirements compared to multimodal transformers
- Proven performance in text-only NER tasks

Due to limited resources and our intention to isolate OCR effects, multimodal models were deferred to future work.

E. Related Work

The extraction of financial data from receipts is an area that has seen considerable growth due to the increasing need for automation in processing digital documents. [5] Various methods, particularly those relying on Optical Character Recognition (OCR) and Natural Language Processing (NLP), have been explored to tackle the challenges in receipt data extraction. One approach that gained attention was developed by **Ghosal et al. (2020)**, where they built a system that combined OCR with traditional machine learning techniques to extract data from receipts. [6] While this method achieved some success, it struggled with inconsistencies in receipt formats and errors caused by low-quality images, which are common in real-world scenarios.

Another significant contribution came from **Xu et al. (2021)**, who took a different route by integrating deep learning models, particularly Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks.[7] These models were

able to capture both the spatial arrangement and the sequential nature of the text on receipts. Their work demonstrated that deep learning approaches can improve the accuracy of metadata extraction, particularly when addressing various receipt formats and fonts.[8] However, issues like dealing with distorted text or low-resolution images still pose challenges. Building on this, **Zhao et al. (2022)** proposed a hybrid approach, combining OCR with NLP techniques such as Named Entity Recognition (NER) to identify key financial entities—like the date, amounts, and vendor names—on receipts. By using semantic analysis, their model was better able to understand the context of the text.. Although this approach showed promise, it still struggled with certain ambiguities in the data, especially when receipts contained unusual formats or unclear handwriting. [9] In a more recent study, Singh and Agarwal (2023) introduced a multi-modal deep learning framework that went beyond simple text extraction. By combining visual elements with textual analysis, their model better handled variations in receipt images, such as distortions and low-quality scans. This multi-modal approach[10] showed significant improvement in accuracy, but challenges such as real-time processing and dataset size remain. Despite these advancements, several challenges remain in the field, including the need for models that can handle noisy receipt images, inconsistent formats, and various languages. While deep learning models have shown great potential, their dependence on large labeled datasets and considerable computational resources limits their practical application in real-world scenarios. The approach presented in this paper aims to address these gaps by utilizing NLP-based methods that focus on understanding the semantic meaning behind financial data. By building a more context-aware model, we hope to improve the accuracy and efficiency of extracting meaningful financial metadata from receipts, moving beyond the limitations of prior methods

IV. RESULTS AND DISCUSSION

A. Results

The following Table shows that PaddleOCR outperforms both PyTesseract and EasyOCR in all metrics, achieving the highest F1-score of 0.958. This indicates that it provides the most accurate entity extraction when paired with the RoBERTa model.

Metric	PyTesseract	EasyOCR	PaddleOCR
Evaluation Loss	0.359	0.177	0.122
Precision	0.896	0.938	0.959
Recall	0.893	0.938	0.957
F1-Score	0.894	0.932	0.958

Table 1: OCR engine comparison on downstream entity extraction accuracy.

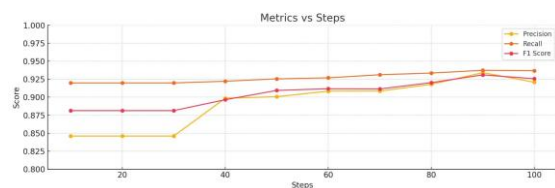


Fig 3: metrics vs step

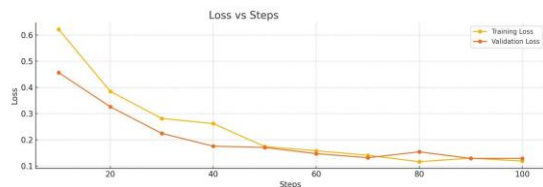


Fig 4: Model training loss across epochs for PaddleOCR-based dataset.



Fig 5: loss vs steps

These numbers highlight [11] PaddleOCR's superiority, especially in accurately recognizing entities when combined with post-OCR token classification. While EasyOCR is a strong contender, PyTesseract struggled with noisy or skewed inputs, even after preprocessing. This evaluation method—benchmarking OCR based on downstream NER—offers a real-world relevance often overlooked in traditional accuracy assessments.

B. Runtime and Performance Analysis

The runtime efficiency of the pipeline was evaluated on a system with a 16 GB VRAM GPU and a CPU with 8GB RAM(Google Colab)

Stage	Avg Time per Document	Bottleneck
OCR (PyTesseract/ EasyOCR/ PaddleOCR)	600-800 ms	OCR engine
Preprocessing	100-200 ms	Image enhancement
Tokenization & RoBERTa inference	250-300 ms	Transformer forward pass
Total	1.1-1.3 sec per document	OCR speed (CPU-bound)

Table 2: Runtime Performance and Bottlenecks of the Document Processing Pipeline

1. **Training Time:** ~2.0 hours for 10 epochs on 626 documents.
2. **Memory Footprint:** ~6–8 GB during training
3. **Batch Size:** 16; learning rate: 3e-5

Although PaddleOCR + RoBERTa delivered the best results (F1: 0.958), it also had slightly higher memory usage.[12] Inference times remain suitable for near real-time applications with further optimization (e.g., GPU-based OCR or quantized models).

V. CONCLUSION AND FUTURE WORK

While the current framework achieves reliable extraction of high-level financial metadata, it can be extended in several directions to support broader applications and improve system performance. One promising extension is the extraction and classification of line-level item information from receipts.[13] This involves parsing individual entries such as product names, quantities, and prices, which can then be categorized into specific domains

(e.g., groceries, electronics, etc.). Such a system would facilitate more granular analysis, including inventory tracking and expense breakdowns. Another important area of future research is multilingual document processing. Many receipts, especially in regions like India, contain a mix of English and regional languages. The proposed system can be expanded to handle multilingual input by integrating language detection and incorporating regional language models. Enhancements to the image preprocessing pipeline are also planned. Using neural-based techniques such as image super-resolution and denoising can improve the clarity of poor-quality receipts before OCR, thereby improving the accuracy of metadata extraction.[14] In addition, mobile deployment is a viable direction for this work. By optimizing the system for resource-constrained devices, real-time extraction of receipt data on smartphones could become possible, enabling broader access and usability in both personal and business contexts. The only issue with implementing this model is getting a proper dataset, which is hard to find. Finally, integrating the extracted metadata into a structured, searchable database would support functionalities such as filtering by date, vendor, or product, and generating summaries or analytics over time.

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